

Planning and Managing the Project: Effort Estimation

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Effort Estimation

- Estimating project costs is one of the crucial aspects of project planning and management
- Estimating cost has to be done as early as possible during the project life cycle

Types of Costs

- Facilities: hardware, space, furniture, telephone, etc.
- Software tools for designing software
- Staff (effort): the biggest component of cost

Software Project Estimation

- S/W cost and effort estimation will never be an exact science
 - Too many variables
 - Human
 - Technical
 - Environmental
 - Political

Pre-requisite to Estimating Effort

- Understand the scope of the project to be built
- Generate an estimate of size

Estimating Size

- The accuracy of a s/w project estimate is predicated on a number of things:
 - The degree to which the planner has properly estimated the size of the product to be built
 - The ability to translate the size estimate into human effort, calendar time, and dollars (required availability of past records)
 - The degree to which the project plan reflects the abilities of the s/w team
 - The stability of product requirements and the environment that supports the s/w engineering effort

Estimating Size

- Sizing represents the project planner's first major challenge
- ▣ *Size* refers to a quantifiable outcome of the s/w project (e.g. LOC and/or FP)

Estimating Size and Effort

- Example of baseline productivity metrics are LOC/pm or FP/pm

An Example

Software Module	Estimated LOC
User interface and control facilities (UICF)	2,300
Two-dimensional geometric analysis (2DGA)	5,300
Three-dimensional geometric analysis (3DGA)	6,800
Database management (DBM)	3,350
Computer graphics display facilities (CGDF)	4,950
Peripheral control function (PCF)	2,100
Design analysis modules (DAM)	8,400
<i>Estimated lines of code</i>	<i>33,200</i>

Effort?

Productivity?

An Example of LOC-Based Estimation

- Estimated lines of code = $W = 33,200$
- Let,
 - Average productivity = $X = 50 \text{ LOC/pd}$
 - Labor rate = $Y = \$100 \text{ per day}$
- So,
 - Cost per line of code = $Z = Y/X = \$2$
 - Total estimated project cost = $W*Z = \$66,400$
 - Estimated effort = $W/X = 664 \text{ person-days}$

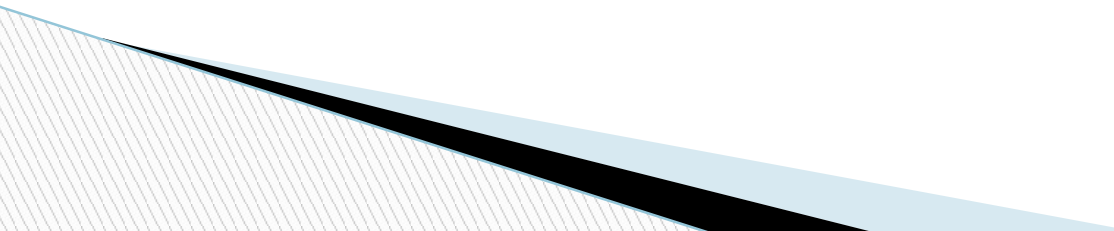
An Example of LOC-Based Estimation

- Estimated lines of code = $W = 33,200$
- Let,
 - Average productivity = $X = 600 \text{ LOC/pm}$
 - Labor rate = $Y = \$5000 \text{ per month}$
- So,
 - Cost per line of code = $Z = Y/X = ?$
 - Total estimated project cost = $W*Z = ?$
 - Estimated effort = $W/X = ?$

3.3 Effort Estimation

Sidebar 3.3 Causes of Inaccurate Estimates

□ Key causes

- Frequent request for change by users
 - Overlooked tasks
 - User's lack of understanding of the requirements
 - Insufficient analysis when developing estimates
 - Lack of coordination of system development, technical services, operations, data administration, and other functions during development
 - Lack of an adequate method or guidelines for estimating
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Decomposition Techniques

- Two different points of view for the decomposition approach
 - Decomposition of the problem
 - Decomposition of the process
- But first, the project planner must
 - Understand the scope of the s/w to be built
 - Generate an estimate of its “size”

Estimating Size

- A hierarchy of estimation models
- Address the following areas
 - Application composition model
 - Early design stage model
 - Post-architecture stage model
- Three different sizing options are available
 - Application points
 - Function points
 - Lines of source code

Problem-Based Estimation of Size

- Example of baseline productivity metrics are LOC/pm or FP/pm
- Making the use of single baseline productivity metric is discouraged
- In general, LOC/pm or FP/pm averages should be computed by project domain
- Local domain averages should be used

Introduction to Function Points

- Can be used effectively as a means for measuring the functionality delivered by a system
- Using historical data, function points can be used to
 - Estimate the cost or effort required to design, code, and test the software
 - Predict the number of errors that will be encountered during testing
 - Forecast the number of components and/or the number of projected source code lines in the implemented system
- Derived using an empirical relationship based on
 - 1) Countable (direct) measures of the software's information domain
 - 2) Assessments of the software's complexity

$$FP = \text{count total} * [0.65 + 0.01 * \text{sum}(F_i)]$$

Introduction to Function Points

□ Information Domain Values

- Number of external inputs
- Number of external outputs
- Number of external inquiries
- Number of internal logical files
- Number of external interface files

□ Value Adjustment Factors (VAF)

- 14 Factors

Information Domain Values

- Number of external inputs
 - Each external input originates from a user or is transmitted from another application
 - They provide distinct application-oriented data or control information
 - They are often used to update internal logical files
 - They are not inquiries (those are counted under another category)
- Number of external outputs
 - Each external output is derived within the application and provides information to the user
 - This refers to reports, screens, error messages, etc.
 - Individual data items within a report or screen are not counted separately

Information Domain Values (continued)

- Number of external inquiries
 - An external inquiry is defined as an online input that results in the generation of some immediate software response
 - The response is in the form of an on-line output
- Number of internal logical files
 - Each internal logical file is a logical grouping of data that resides within the application's boundary and is maintained via external inputs
- Number of external interface files
 - Each external interface file is a logical grouping of data that resides external to the application but provides data that may be of use to the application

Value Adjustment Factors

- 1) Does the system require reliable backup and recovery?
- 2) Are specialized data communications required to transfer information to or from the application?
- 3) Are there distributed processing functions?
- 4) Is performance critical?
- 5) Will the system run in an existing, heavily utilized operational environment?
- 6) Does the system require on-line data entry?
- 7) Does the on-line data entry require the input transaction to be built over multiple screens or operations?

Value Adjustment Factors (continued)

- 8) Are the internal logical files updated on-line?
- 9) Are the inputs, outputs, files, or inquiries complex?
- 10) Is the internal processing complex?
- 11) Is the code designed to be reusable?
- 12) Are conversion and installation included in the design?
- 13) Is the system designed for multiple installations in different organizations?
- 14) Is the application designed to facilitate change and for ease of use by the user?

Function Point Computation

- 1) Identify/collect the information domain values
- 2) Complete the table shown below to get the count total (aka unadjusted function points)
 - Associate a weighting factor (i.e., complexity value) with each count based on criteria established by the software development organization
- 3) Evaluate and sum up the adjustment factors (see example in the next few slides)
 - “F_i” refers to 14 value adjustment factors, with each ranging in value from 0 (not important) to 5 (absolutely essential)
- 4) Compute the number of function points (FP) aka Adjusted FP

$$FP = \text{count total} * [0.65 + 0.01 * \text{sum}(F_i)]$$

Information		Weighting Factor			
<u>Domain Value</u>	<u>Count</u>	<u>Simple</u>	<u>Average</u>	<u>Complex</u>	
External Inputs	_____ x	3	4	6 =	_____
External Outputs	_____ x	4	5	7 =	_____
External Inquiries	_____ x	3	4	6 =	_____
Internal Logical Files	_____ x	7	10	15 =	_____
External Interface Files	_____ x	5	7	10 =	_____
Count total					_____

Function Point Example

Information		Weighting Factor				
<u>Domain Value</u>	<u>Count</u>		<u>Simple</u>	<u>Average</u>	<u>Complex</u>	
External Inputs	3	x	3	4	6	= 9
External Outputs	2	x	4	5	7	= 8
External Inquiries	2	x	3	4	6	= 6
Internal Logical Files	1	x	7	10	15	= 7
External Interface Files	4	x	5	7	10	= 20
Count total				50		

- $FP = \text{count total} * [0.65 + 0.01 * \sum(F_i)]$
- $FP = 50 * [0.65 + (0.01 * 46)]$
- $FP = 55.5$ (rounded up to 56)

An Example of FP-Based Estimation

Information Domain Value	Count		Weighting factor				
			Simple	Average	Complex		
External Inputs (EIS)	6	X	3	4	6	=	
External Outputs (EOs)	3	X	4	5	7	=	
External Inquiries (EQs)	5	X	3	4	6	=	
Internal Logical Files (ILFs)	2	X	7	10	15	=	
External Interface Files (EIFs)	4	X	5	7	10	=	
Count Total							

Computing function points

An Example of FP-Based Estimation

Factor	Value
1. Backup and recovery	4
2. Data communications	2
3. Distributed processing	0
4. Performance critical	4
5. Existing operating environment	3
6. On-line data entry	4
7. Input transaction over multiple screens	5
8. ILFs updated online	3
9. Information domain values complex	5
10. Internal processing complex	5
11. Code designed for reuse	4
12. Conversion/installation in design	3
13. Multiple installations	5
14. Application designed for change	5

Value Adjustment Factors

An Example of FP-Based Estimation

□ Now,

- $FP_{\text{estimated}} = \text{count_total} \times [0.65 + 0.01 \times \sum (F_i)]$
 - F_i ($i = 1$ to 14 are value adjustment factors)

□ So,

- $FP_{\text{estimated}} = W = 320 \times [0.65 + 0.01 \times 52] = 375$
(approx.)

□ Let,

- Average Productivity = $X = 6.5$ FP/pm
- Labor rate = $Y = \$8,000$ per month

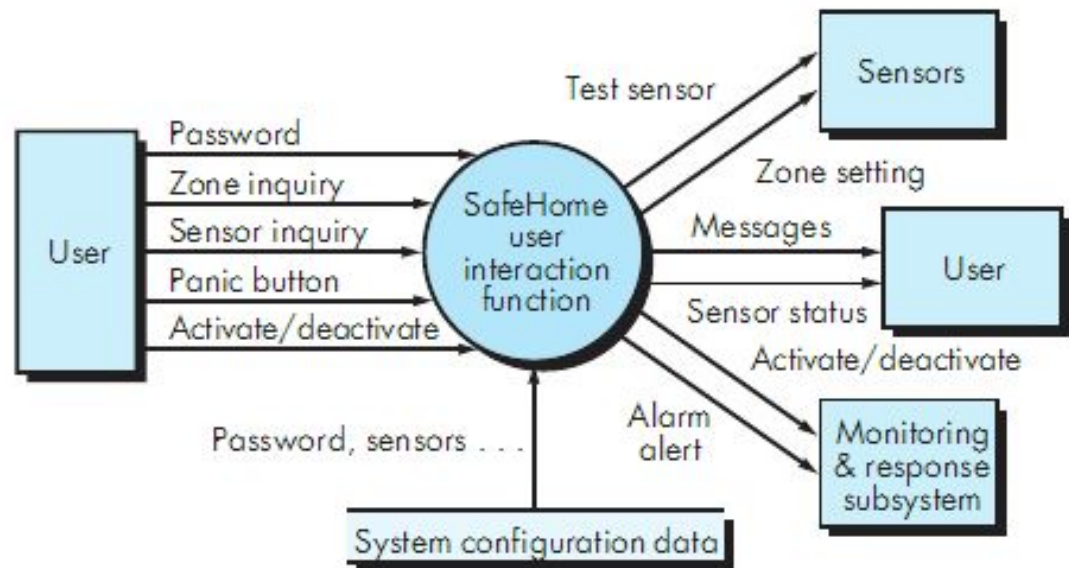
□ So,

- Cost per FP = $Z = Y/X = \$1,230$ (approx.)
- Total estimated project cost = $W \times Z = \$461,000$ (approx.)
- Estimated effort = $W/X = 58$ person months (approx)

An Example

- password, panic button, activate/deactivate
- zone inquiry, sensor inquiry
- system configuration file
- messages, sensor status
- test sensor, zone setting, alarm alert, a/d

- Let $\Sigma F_i = 46$
- Size in FP?
- Effort?
- Cost?



Effort Estimation

COCOMO II: Stages of Development

- Application composition
 - prototyping to resolve high-risk user interface issues
 - size estimates in application/object points
- Early design
 - to explore alternative architectures and concepts
 - size estimates in function points
- Post architecture
 - development has begun
 - size estimates in lines of code

Estimating Size

- Application points are computed using counts of the number of
 - Screens (at the user interface)
 - Reports
 - Components likely to be required to build the application

Estimating Size

Object type	Complexity weight		
	Simple	Medium	Difficult
Screen	1	2	3
Report	2	5	8
Components			10

Complexity weights for object types

Estimating Size

Object type	Complexity weight		
	Simple	Medium	Difficult
Screen	1	2	3
Report	2	5	8
Components			10

Determine Object Points: Add all weighted object instances to get object point count

Estimating Size

- ✍ Determine object points based on reuse of code
- ▶ New Object Points (NOP) =
$$\frac{\text{Object points} * (100 - \% \text{ reuse})}{100}$$
- ▶ NOP differ from the object point count because there may be reuse

Estimating Size

Developer's experience/capability	Very low	Low	Nominal	High	Very High
Environment maturity/capability	Very low	Low	Nominal	High	Very High
PROD	4	7	13	25	50

Productivity rate for object points

Estimating Size

$$\text{Effort in PM} = \frac{NOP}{PROD}$$

Cocomo-II Example

- Use Cocomo-II model to estimate effort required to build an Order Taking system for restaurant that produces 15 screens, 6

Object	Count	Complexity	Weight Factor	Total Objects
Screen	15	Medium	2	30
Report	6	Medium	5	30
Components	0	NA	NA	0
Total Object Points:				60

Cocomo-II Example

- It is mentioned that 70% new software components will be developed. So, the remaining 30% can be reused.

New Object Points (NOP) =

$$\frac{\text{Object points} * (100 - \% \text{ reuse})}{100}$$

$$\text{NOP} = \frac{60 * (100 - 30)}{100}$$

$$\text{NOP} = \frac{60 * 70}{100}$$

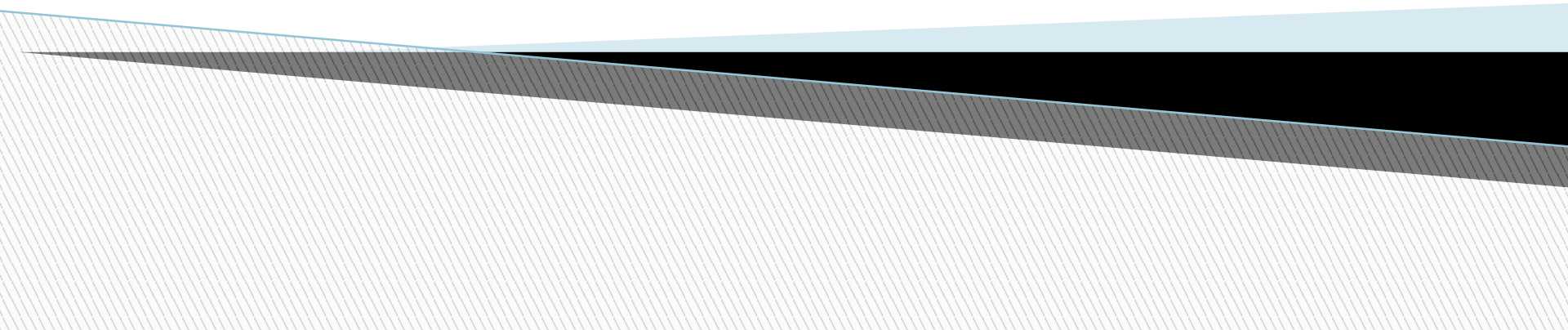
$$\text{NOP} = 42$$

Cocomo-II Example

- ✍ Since developer/environment maturity is average, productivity (PROD) = 13
- ▶ $\text{Effort} = \frac{NOP}{PROD}$
- ▶ $\text{Effort} = \frac{42}{13}$
- ▶ $\text{Effort} = 3.23$ person months

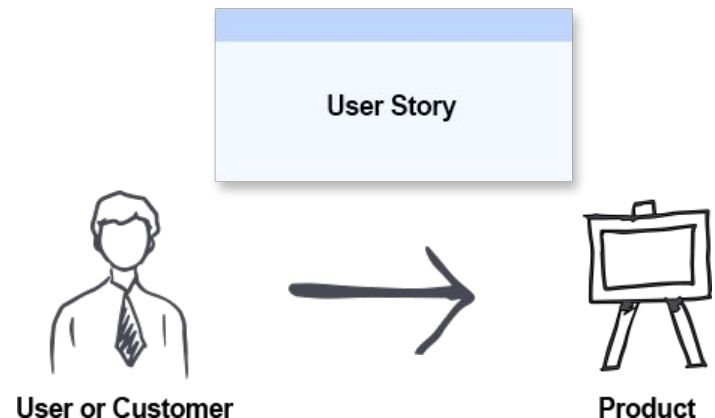
Management Activities in SE

Estimation in Agile Development (Recap)
Discussed before Mid 1



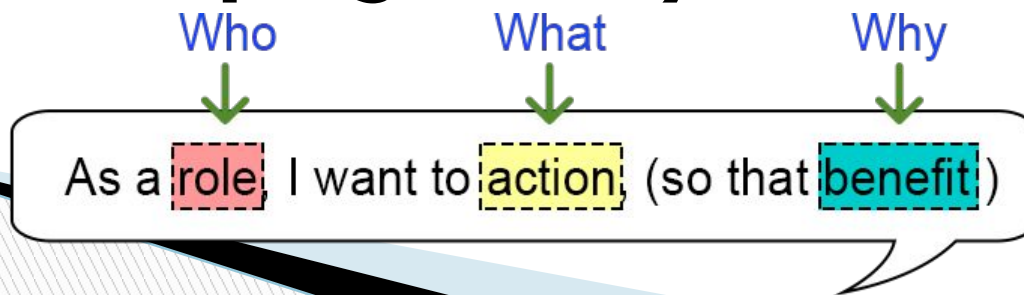
Estimation in Agile

- ❑ Project Velocity
 - The amount of work an agile team can complete in a sprint
- ❑ Story Points
 - The amount of effort required to implement a user story
- ❑ User Story
 - A requirement or general explanation of software feature



Examples of User Stories

- As a **[customer]**, I want **[shopping cart feature]** so that **[I can easily purchase items online]**.
- As an **[manager]**, I want to **[generate a report]** so that **[I can understand which departments need more resources]**.
- As a **[customer]**, I want to **[receive an SMS when the item is arrived]** so that **[I can go pick it up right away]**



As who, I want what
so that why



Story Map

Estimate & Spike

Sprint

Product Backlog

General Activity

Find Job

Manage
VacancyRecruit
CandidateGeneral User
Task

Browse Jobs

Post Resume

Receive Job
Alert

Post Vacancy

Amend
VacancyCancel
VacancySearch
CandidatesContact
Candidates

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As a system
administrator, I
want to obtainAs a job seeker,
I want to
search jobsAs a job seeker,
I want to
upload myAs a job seeker,
I want to
subscribe forAs a employer,
I want to
submit a jobAs a employer,
I want to edit a
vacancyAs a employer,
I want to
remove aAs a job seeker,
I want to see
the job seekersAs a employer,
I want to send
a privateAs a system
administrator, I
want to obtainAs a job seeker,
I want to select
desiredAs a job seeker,
I want to
upload myAs a employer,
I want to
register as anAs a employer,
I want to
receive emailAs a job seeker,
I want to read
privateAs a job seeker,
I want to
search jobs byAs a job seeker,
I want to
upload myAs a
department
staff, I want toAs a job seeker,
I want to
remove private

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As a system
administrator, I
want to keepAs a job seeker,
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search jobs byAs a job seeker,
I want to read
the privacyAs a job seeker,
I want to
receive SMSAs a
department
staff, I want toAs a
department
staff, I want toAs a employer,
I want to
receive SMSAs a employer,
I want to
searchAs a job seeker,
I want to
archive a

As a job seeker,

As a

As a

As a employer,

Configuration



Estimation in Agile

- Project Velocity
 - $\text{Story points completed} \div \text{Number of sprints}$

References

- SE by Pfleeger
- SE by Pressman
- UCF slides for SE course